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Supplementary material

# APPENDIX

Appendix to: Krishnamoorthi R, Bomman S, Benias P, et al. Efficacy and safety of endoscopic duodenal stent versus endoscopic or surgical gastrojejunostomy to treat malignant gastric outlet obstruction: systematic review and meta-analysis

**eTable 1**. Search strategy.

Database Used: Embase and MEDLINE (via Embase.com) January 2015 to February 16, 2021
**Accessed:** February 17, 2021

| No. | Query |
|---|---|
| #13 | #9 OR #11 OR #12 |
| #12 | #11 NOT ([conference abstract]/lim OR [conference paper]/lim OR [conference review]/lim OR [data papers]/lim OR [editorial]/lim OR [erratum]/lim OR [letter]/lim OR [note]/lim OR [short survey]/lim OR 'case report'/de) |
| #11 | 'axios' AND ('boston scientific' OR 'boston sci' OR bsci OR bsc) AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |
| #10 | #7 NOT ([conference abstract]/lim OR [conference paper]/lim OR [conference review]/lim OR [data papers]/lim OR [editorial]/lim OR [erratum]/lim OR [letter]/lim OR [note]/lim OR [short survey]/lim OR 'case report'/de) |
| #9 | #8 NOT ([conference abstract]/lim OR [conference paper]/lim OR [conference review]/lim OR [data papers]/lim OR [editorial]/lim OR [erratum]/lim OR [letter]/lim OR [note]/lim OR [short survey]/lim OR 'case report'/de) |
| #8 | (#1 OR #2) AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |
| #7 | (#5 OR #6) AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |
| #6 | #3 AND axios AND ('boston scientific' OR 'boston sci' OR bsci OR bsc) |
| #5 | (#1 OR #2) AND axios AND ('boston scientific' OR 'boston sci' OR bsci OR bsc) |
| #4 | #1 OR #2 |
| #3 | ('endoscopic gastrojejunostomy' OR ('bypass surgery' NEAR/5 'gastric outlet obstruction')) AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |
| #2 | ('malignant gastric outlet obstruction' OR ('gastric outlet obstruction' NEAR/10 malignant) OR 'malignant gastroduodenal obstruction' OR 'malignant duodenal obstruction' OR 'malignant pyloric obstruction' OR ('gastric outlet obstruction' NEAR/10 benign) OR 'benign gastric outlet obstruction') AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |
| #1 | ('lumen apposing metal stent*' OR 'lumen apposing metallic stent*' OR 'lumen apposing stent*' OR ('self expandable metallic stent*' NEAR/5 duodenal) OR ('self expandable metal stent*' NEAR/5 duodenal)) AND [english]/lim AND [humans]/lim AND ([clinical study]/lim OR 'systematic review*':ti) AND [2-7-2020]/sd NOT [17-2-2021]/sd |



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## Supplementary material

**eTable 2**. Assessment of methodologic quality of studies utilizing a modified Newcastle-Ottawa Scale.

**Modified Newcastle-Ottawa Scale questions**

1. Did the patient(s) represent the whole case(s) of the medical center? Cases included represented the general population of gastric outlet obstruction
2. Was the diagnosis correctly made? Were only malignant patients included?
3. Was follow-up long enough for outcomes to occur? Reported adequate follow-up time.
4. Were all important data cited in the report? Reported clinical success and at least 2 outcomes (NR = did not define “clinical success,” but study could still score a “Yes” if they reported (undefined) clinical success rate)
5. Was the outcome correctly ascertained? Provided definition of clinical success.

| Study | Year | Question 1 | | Question 2 | | Question 3 | | Question 4 | | Question 5 | | Methodologic Quality |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Yes | No | Yes | No | Yes | No | Yes | No | Yes | No | |
| Chiu[1] | 2015 | X | | X | | X | | X | | X | | Good |
| JW Kim[2] | 2015 | X | | X | | X | | X | | X | | Good |
| SH Kim[3] | 2015 | X | | X | | X | | X | | X | | Good |
| H Lee[4] | 2015 | | X (gastric cancer only) | X | | X | | X | | X | | Moderate |
| JE Lee[5] | 2015 | X | | X | | X | | X | | X | | Good |
| D Oh[6] | 2015 | X | | X | | X | | X | | X | | Good |
| SY Oh[7] | 2015 | X | | X | | X | | X | | X | | Good |
| Park[8] | 2015 | | X (gastric cancer only) | X | | X | | X | | X | | Moderate |
| Sato[9] | 2015 | X | | X | | X | | X | | | NR | Moderate |
| Trotter[10] | 2015 | X | | X | | X | | X | | X | | Good |
| Fiori[11] | 2016 | | X (metastatic adenocarcinoma of antropyloric region only) | X | | X | | X | | | NR | Poor |
| Grunwald[12] | 2016 | X | | X | | X | | X | | X | | Good |
| Itoi[13] | 2016 | X | | X | | X | | | X (few AEs specified) | | NR | Poor |
| Jung[14] | 2016 | X | | X | | X | | X | | X | | Good |
| Kato[15] | 2016 | X | | X | | X | | X | | X | | Good |
| Khan[16] | 2016 | X | | X | | X | | X (barely) | | X | | Good |
| Kobayashi[17] | 2016 | | X (pancreatic cancer only) | X | | X | | X | | X | | Moderate |
| Lye[18] | 2016 | X | | X | | X | | X | | X | | Good |
| Okuwaki[19] | 2016 | | X (advanced pancreatic | X | | X | | X | | X | | Moderate |



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| Study | Year | Question 1 | | Question 2 | | Question 3 | | Question 4 | | Question 5 | | Methodologic Quality |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Yes | No | Yes | No | Yes | No | Yes | No | Yes | No | |
| | | | cancer or advanced biliary cancer only) | | | | | | | | | |
| J-H Park (1)[20] | 2016 | X | | X | | X | | X | | | NR | Moderate |
| J-H Park (2)[21] | 2016 | | X (gastric cancer only | X | | X | | X | | X | | Moderate |
| Rademacher[22] | 2016 | X | | X | | X | | X | | X | | Good |
| Sasaki[23] | 2016 | X | | X | | X | | X | | X | | Good |
| Shin[24] | 2016 | X | | X | | X | | X | | X | | Good |
| Tsauo[25] | 2016 | | X (pancreatic cancer only) | X | | X | | X | | X | | Moderate |
| Yamao[26] | 2016 | X | | X | | X | | X | | X | | Good |
| Bulut[27] | 2017 | X | | X | | X | | X | | X | | Good |
| Chen[28] | 2017 | X | | X | | X | | X | | X | | Good |
| Hori[29] | 2017 | X | | X | | X | | X | | X | | Good |
| Jang[30] | 2017 | | X (gastric cancer only) | X | | | X (patients excluded for follow-up < 1 month) | X | | X | | Poor |
| Khashab[31] | 2017 | X | | X | | X | | X | | X | | Good |
| Kim[32] | 2017 | | X (periampullary ca only) | X | | X | | X | | X | | Moderate |
| Ojima[33] | 2017 | | X (gastric cancer only) | X | | X | | X | | | NR | Poor |
| Perez-Miranda[34] | 2017 | X | | | X | X | | X | | X | | Moderate |
| Takahara[35] | 2017 | X | | X | | X | | X | | X | | Good |
| Tanaka[36] | 2017 | | X (gastric cancer only) | X | | X | | X | | | NR | Poor |
| Tsauo[37] | 2017 | X | | X | | X | | X | | X | | Good |
| Ye[38] | 2017 | X | | X | | X | | X | | X | | Good |
| Yoshida[39] | 2017 | | X (pancreatic cancer only) | X | | X | | X | | X | | Moderate |
| Bekheet[40] | 2018 | X | | X | | X | | X | | X | | Good |
| Choi[41] | 2018 | X | | X | | X | | X | | X | | Good |
| Leiyuan[42] | 2018 | X | | X | | X | | X | | | NR | Moderate |
| Uemura[43] | 2018 | | X (pancreatic cancer only) | X | | X | | X | | X | | Moderate |
| Yukimoto[44] | 2018 | X | | X | | | NR | X | | X | | Moderate |
| Ge[45] | 2019 | X | | X | | X | | X | | X | | Good |
| Jang[46] | 2019 | X | | X | | X | | X | | X | | Good |



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| Study | Year | Question 1 | | Question 2 | | Question 3 | | Question 4 | | Question 5 | | Methodologic Quality |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Yes | No | Yes | No | Yes | No | Yes | No | Yes | No | |
| Kerdsirichairat[47] | 2019 | X | | X | | | | X | | X | | Good |
| Kumar[48] | 2019 | X | | X | | X | | X | | X | | Good |
| Ramos[49] | 2019 | | X (gastric cancer only) | X | | X | | X | | | NR | Poor |
| Ratone[50] | 2019 | X | | X | | X | | X | | X | | Good |
| Sterpetti[51] | 2019 | X | | X | | X | | X | | | NR | Moderate |
| Alcala-Gonzalez[52] | 2020 | X | | X | | X | | X | | X | | Good |
| Kastelijn[53] | 2020 | X | | X | | X | | X | | X | | Good |
| Miwa[54] | 2020 | X | | X | | X | | X | | X | | Good |
| Mo[55] | 2020 | X | | X | | X | | X | | X | | Good |
| Wu[56] | 2020 | X | | X | | X | | X | | X | | Good |
| Xu[57] | 2020 | X | | X | | X | | X | | X | | Good |
| Yildirim[58] | 2020 | X | | X | | X | | X | | | NR | Moderate |
| Hindryckx[59] | 2021 | X | | | X | X | | X | | X | | Moderate |
| Kouanda[60] | 2021 | X | | X | | X | | X | | X | | Good |
| Yamao[61] | 2021 | X | | X | | X | | X | | X | | Good |



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## Supplementary material

**eTable 3**. Analysis of heterogeneity by outcome and procedure in all studies.

| Procedure | Outcome | # of Studies | # of Patients | $I^2$ (95% CI) |
|---|---|---|---|---|
| Duodenal SEMS | Technical success | 45 | 4413 | 65.6% (52.9% to 74.8%) |
| EUS-GJ | Technical success | 8 | 245 | 67.4% (31.4% to 84.5%) |
| Surgical GJ | Technical success | 13 | 564 | 0.0% (0.0% to 16.4%) |
| Duodenal SEMS | Clinical success | 45 | 4590 | 79.3% (72.7% to 84.2%) |
| EUS-GJ | Clinical success | 8 | 245 | 54.6% (0.0% to 79.5%) |
| Surgical GJ | Clinical success | 13 | 588 | 77.6% (62.1% to 86.8%) |
| Duodenal SEMS | Preprocedural GOOSS score | 27 | 2655 | 97.0% (96.3% to 97.5%) |
| EUS-GJ | Preprocedural GOOSS score | 2 | 65 | 0.0% (., .) |
| Surgical GJ | Preprocedural GOOSS score | 6 | 215 | 67.7% (23.3% to 86.4%) |
| Duodenal SEMS | Postprocedural GOOSS score | 19 | 2184 | 95.8% (94.6% to 96.8%) |
| EUS-GJ | Postprocedural GOOSS score | 2 | 59 | 95.1% (85.5 % to 98.4%) |
| Surgical GJ | Postprocedural GOOSS score | 5 | 180 | 99.0% (98.5% to 99.3%) |
| Duodenal SEMS | Recurrence of GOO | 11 | 573 | 83.4% (71.8% to 90.3%) |
| EUS-GJ | Recurrence of GOO | 1 | 25 | --- |
| Surgical GJ | Recurrence of GOO | 8 | 342 | 48.6% (0.0% to 77.1%) |
| Duodenal SEMS | Reintervention | 33 | 2963 | 81.3% (74.4% to 86.3%) |
| EUS-GJ | Reintervention | 4 | 129 | 44.3% (0.0% to 81.4%) |
| Surgical GJ | Reintervention | 9 | 418 | 75.3% (52.5% to 87.2%) |
| Duodenal SEMS | Procedure Related Complications | 43 | 4285 | 91.6% (89.6% to 93.2%) |
| EUS-GJ | Procedure Related Complications | 6 | 189 | 0.0% (0.0% to 65.4%) |
| Surgical GJ | Procedure Related Complications | 16 | 746 | 67.0% (44.3% to 80.5%) |
| Duodenal SEMS | Bleeding | 25 | 2854 | 65.1% (46.7% to 77.2%) |
| EUS-GJ | Bleeding | 4 | 141 | 57.0% (0.0% to 85.7%) |
| Surgical GJ | Bleeding | 9 | 412 | 0.0% (0.0% to 16.7%) |
| Duodenal SEMS | Perforation | 24 | 2823 | 21.1% (0.0% to 52.2%) |
| EUS-GJ | Perforation | 3 | 105 | 65.7% (0.0% to 90.1%) |
| Surgical GJ | Perforation | 3 | 170 | 47.5% (0.0% to 84.6%) |
| Duodenal SEMS | Stent migration | 33 | 3451 | 60.7% (42.6% to 73.1%) |
| EUS-GJ | Stent migration | 4 | 116 | 69.6% (12.4% to 89.4%) |
| Duodenal SEMS | Stent occlusion | 22 | 1993 | 92.2% (89.5% to 94.2%) |
| EUS-GJ | Stent occlusion | 3 | 69 | 22.6% (0.0% to 92.0%) |
| Duodenal SEMS | Tissue ingrowth | 22 | 2172 | 85.7% (79.7% to 90.0%) |
| EUS-GJ | Tissue ingrowth | 1 | 24 | --- |
| Duodenal SEMS | Stent patency | 3 | 140 | 0.0% (0.0% to 87.6%) |
| Duodenal SEMS | Tissue overgrowth | 20 | 1962 | 45.4% (7.8% to 67.7%) |
| Surgical GJ | Laparoscopic complication | 5 | 196 | 76.8% (43.8% to 90.5%) |
| Duodenal SEMS | Deaths reported in AE section | 26 | 2151 | 70.8% (56.6% to 80.4%) |
| EUS-GJ | Deaths reported in AE section | 4 | 151 | 76.5% (35.7% to 91.4%) |
| Surgical GJ | Deaths reported in AE section | 8 | 421 | 54.6% (0.0% to 79.5%) |



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SEMS self-expanding metal stent  EUS-GJ endoscopic ultrasound-guided gastrojejunostomy
GOO gastric outlet obstruction  GOOSS Gastric Outlet Obstruction Scoring System  AE adverse event



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**eTable 4**. Publication bias testing

| Outcome | Rank correlation test of funnel plot asymmetry [Begg and Mazumdar] | | | Rank correlation test of funnel plot asymmetry (with continuity correction) [Begg and Mazumdar] | | | Linear regression test of funnel plot asymmetry [Eggers] | | |
|---|---|---|---|---|---|---|---|---|---|
| | Duodenal SEMS | EUS-GJ | Surgical GJ | Duodenal SEMS | EUS-GJ | Surgical GJ | Duodenal SEMS | EUS-GJ | Surgical GJ |
| Technical success | 0.0869 | 0.7084 | 0.0016 | 0.0887 | 0.8031 | 0.0020 | 0.6243 | 0.4396 | 0.0051 |
| Clinical success | 0.7394 | 0.1051 | 0.9513 | 0.7468 | 0.1346 | 1.0000 | 0.5886 | 0.1681 | 0.5438 |
| Preprocedural GOOSS score | 0.5455 | 3.0000 | 0.3476 | 0.5594 | 3.0000 | 0.4524 | 0.0006 | 3.0000 | 0.3233 |
| Postprocedural GOOSS score | 0.1955 | 3.0000 | 0.6242 | 0.2079 | 3.0000 | 0.8065 | 0.0517 | 3.0000 | 0.5875 |
| Recurrence of GOO | 0.4835 | 3.0000 | 0.8046 | 0.5334 | 3.0000 | 0.9015 | 0.6230 | 3.0000 | 0.5985 |
| Reintervention | 0.1453 | 0.4969 | 0.1707 | 0.1496 | 0.7341 | 0.2060 | 0.1658 | 0.4270 | 0.0918 |
| Procedural complications | 0.3460 | 0.0195 | 0.0701 | 0.3514 | 0.0323 | 0.0774 | 0.6682 | 0.0671 | 0.0739 |
| Bleeding | 0.3382 | 0.7180 | 0.8348 | 0.3501 | 1.0000 | 0.9170 | 0.0626 | 0.5224 | 0.1233 |
| Perforation | 0.4560 | 0.1172 | 0.6015 | 0.4712 | 0.2963 | 1.0000 | 0.9056 | 0.0992 | 0.8675 |
| Stent migration | 0.7683 | 0.2786 | --- | 0.7801 | 0.4701 | --- | 0.8298 | 0.0709 | --- |
| Stent occlusion | 0.5726 | 0.6015 | --- | 0.5920 | 1.0000 | --- | 0.8978 | 0.5684 | --- |
| Tissue ingrowth | 0.2587 | 3.0000 | --- | 0.2708 | 3.0000 | --- | 0.1399 | 3.0000 | --- |
| Stent patency | 0.1172 | --- | --- | 0.2963 | --- | --- | 0.2048 | --- | --- |
| Tissue overgrowth | 0.0349 | --- | --- | 0.0378 | --- | --- | 0.0445 | --- | --- |
| Laparoscopic complication | --- | --- | 0.6242 | --- | --- | 0.8065 | --- | --- | 0.6699 |
| Deaths reported in AE section | 0.1280 | 1.0000 | 1.0000 | 0.1336 | 1.0000 | 1.0000 | 0.8817 | 0.7039 | 0.7236 |

All numbers are p-values from the rank correlation or linear regression tests for funnel plot symmetry.
* - the test could not be conducted for the EUS-GJ studies because only 1 study was available.



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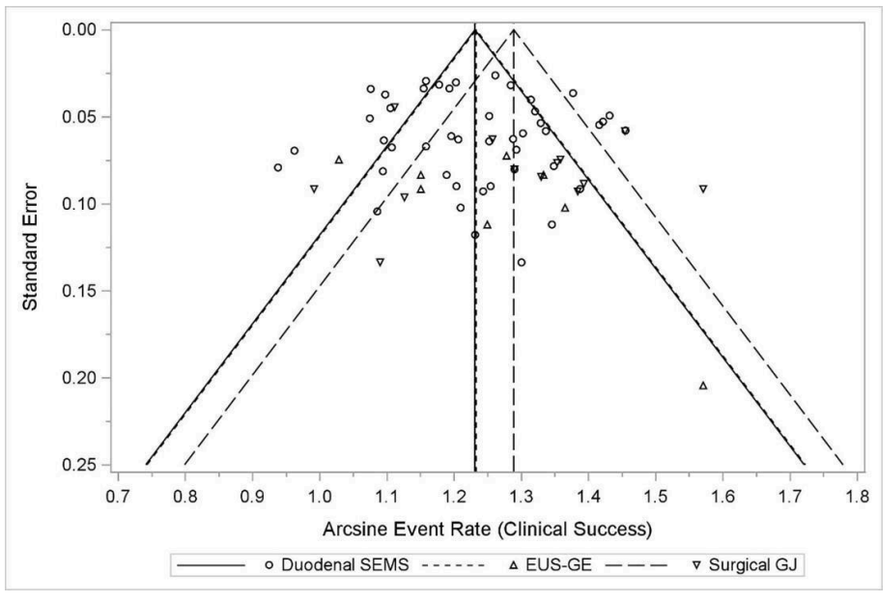
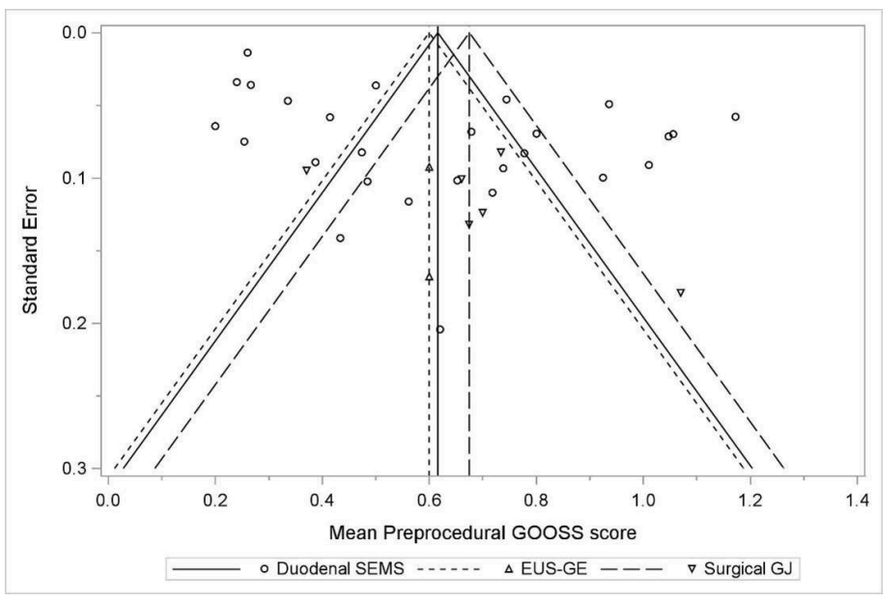
## Supplementary material

**eFigure 1.** Funnel plots for studies of duodenal SEMS compared to EUS-GJ and surgical GJ: efficacy outcomes: (A) clinical success, (B) preprocedural GOOSS score, (C) postprocedural GOOSS score, (D) recurrence of GOO, and (E) reintervention; safety outcomes: (F) bleeding (G) perforation, (H) stent migration, (I) stent occlusion, (J) tissue ingrowth, (K) stent patency, (L) tissue overgrowth, (M) deaths reported in AE section

**A.** Clinical success




**B.** Preprocedural GOOSS score



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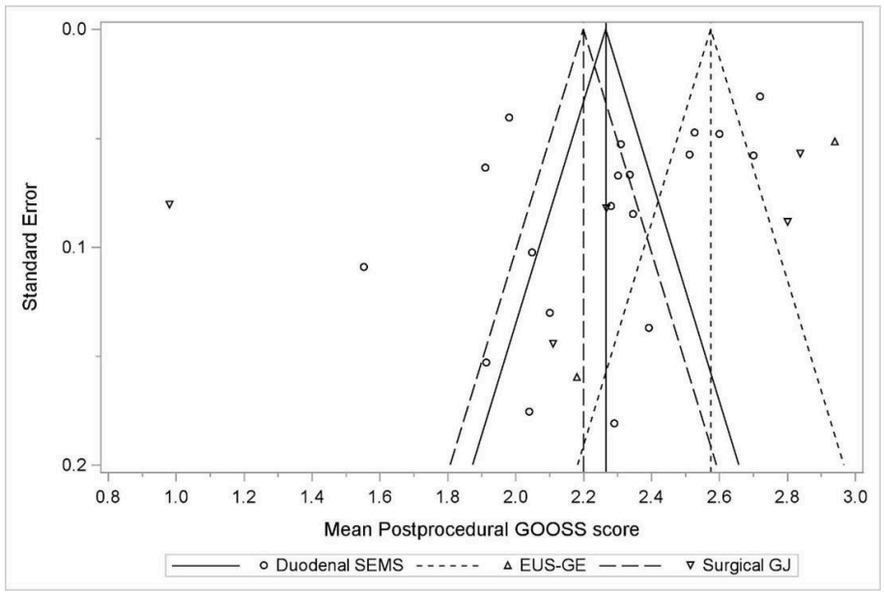
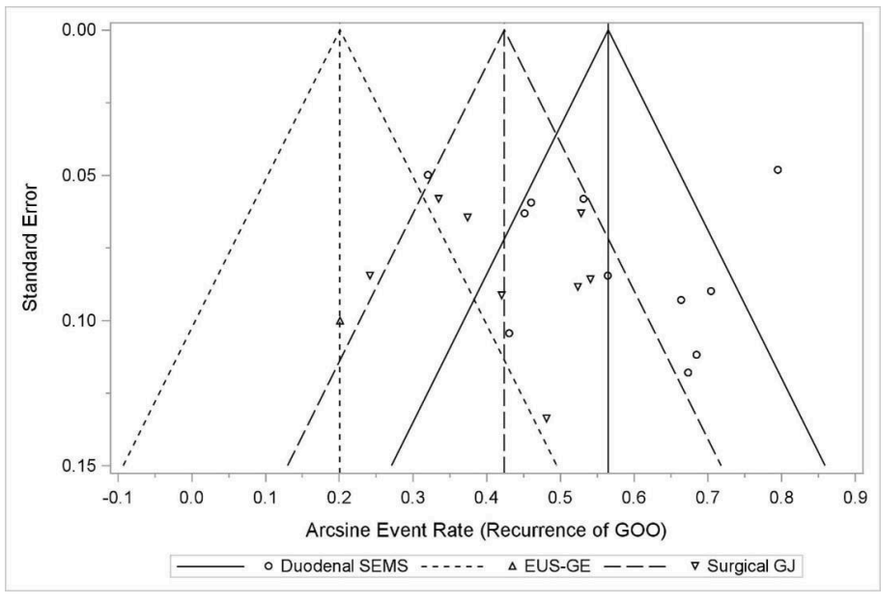
**C.** Postprocedural GOOSS score




**D.** Recurrence of GOO



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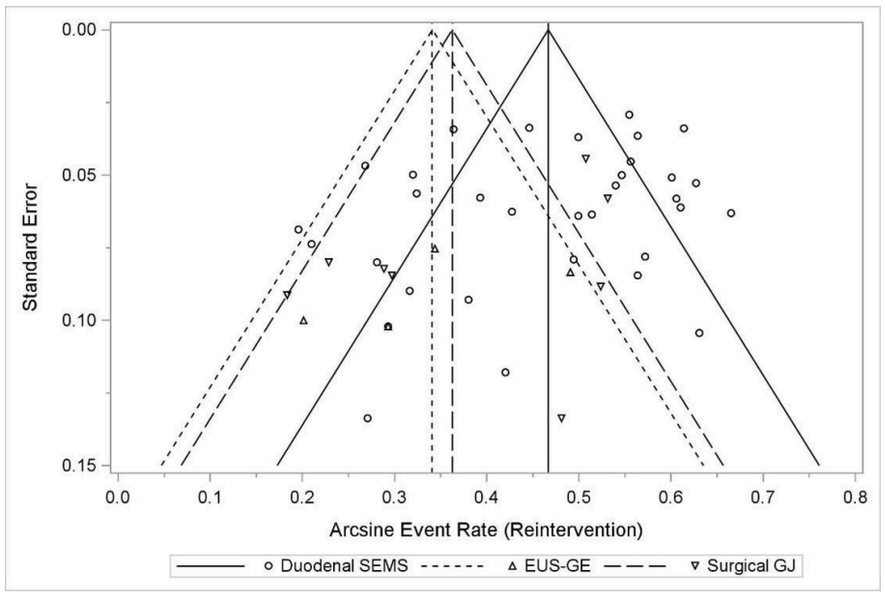
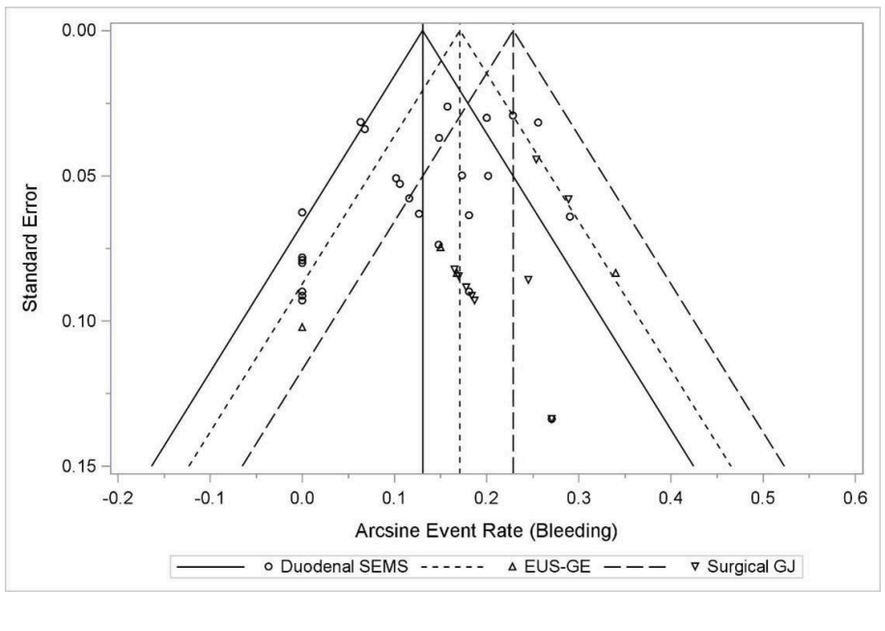
**E.** Reintervention




**F.** Bleeding



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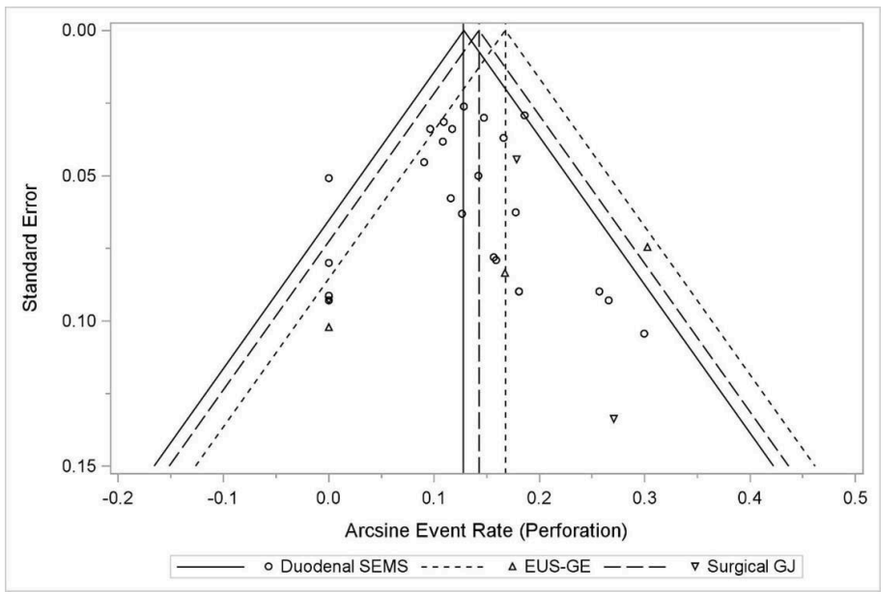
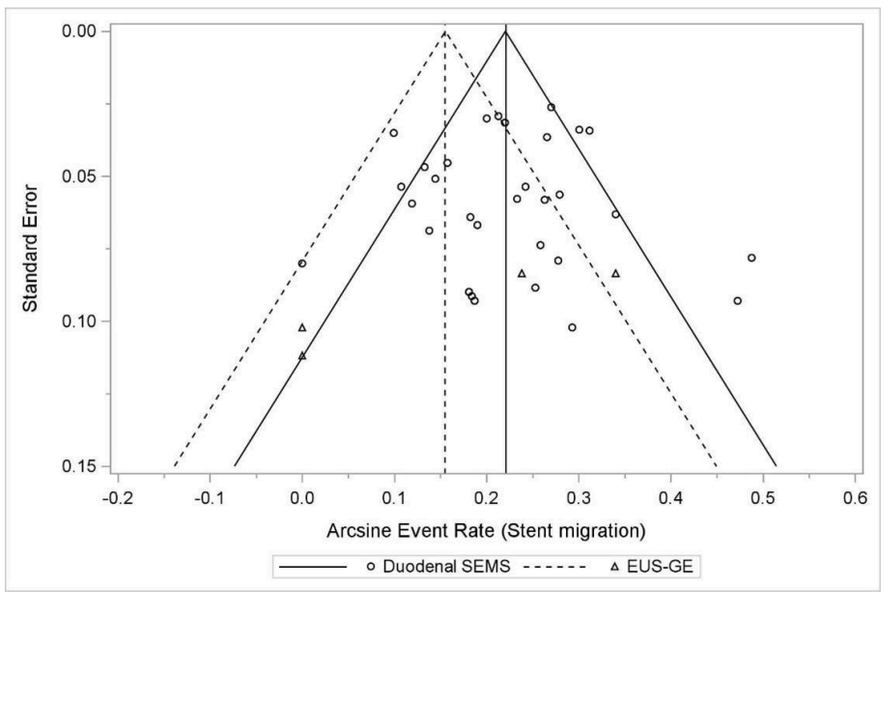
**G.** Perforation




**H.** Stent migration



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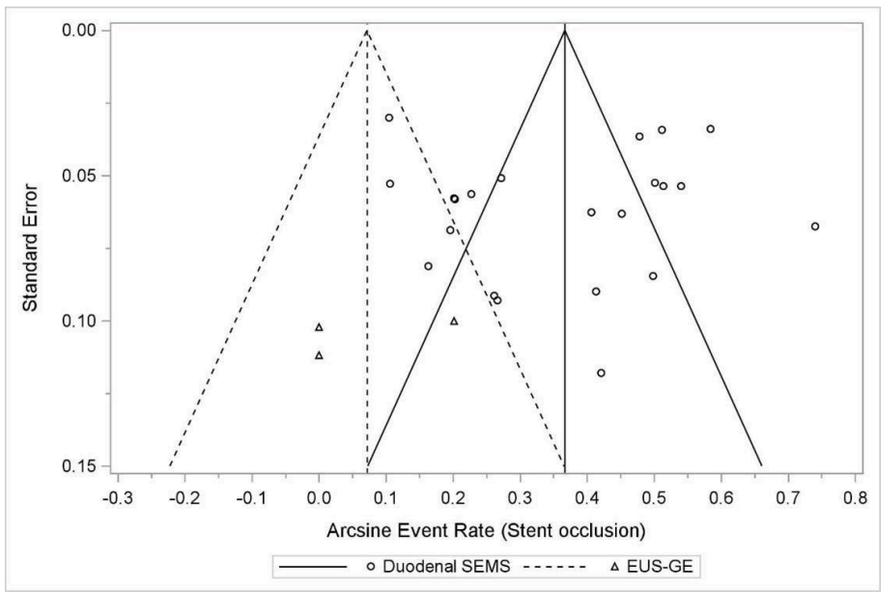
**I.** Stent occlusion




**J.** Tissue ingrowth



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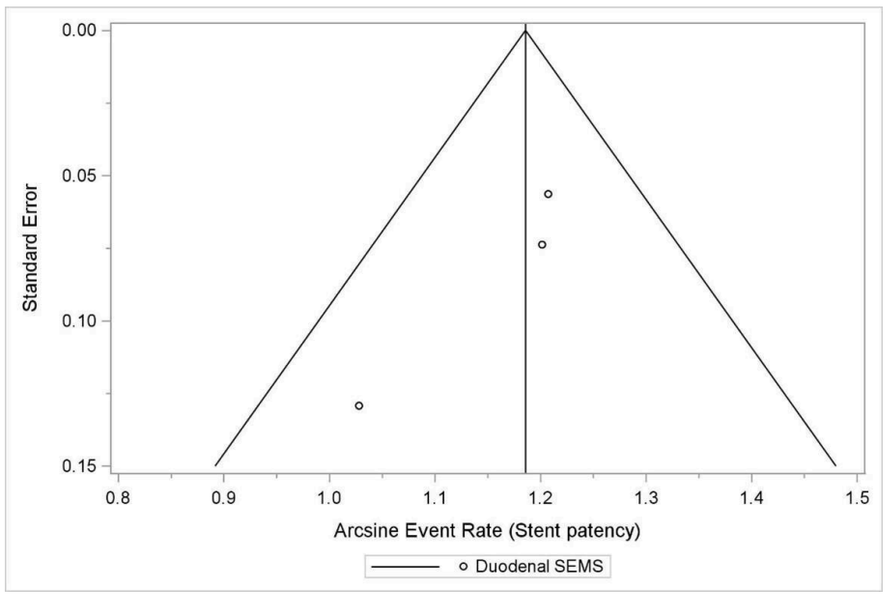
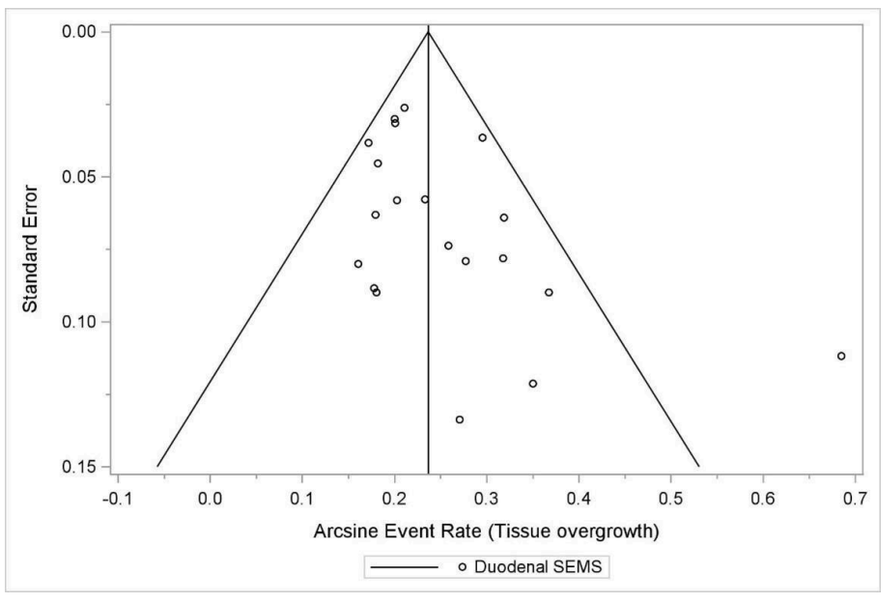
**K.** Stent patency




**L.** Tissue overgrowth



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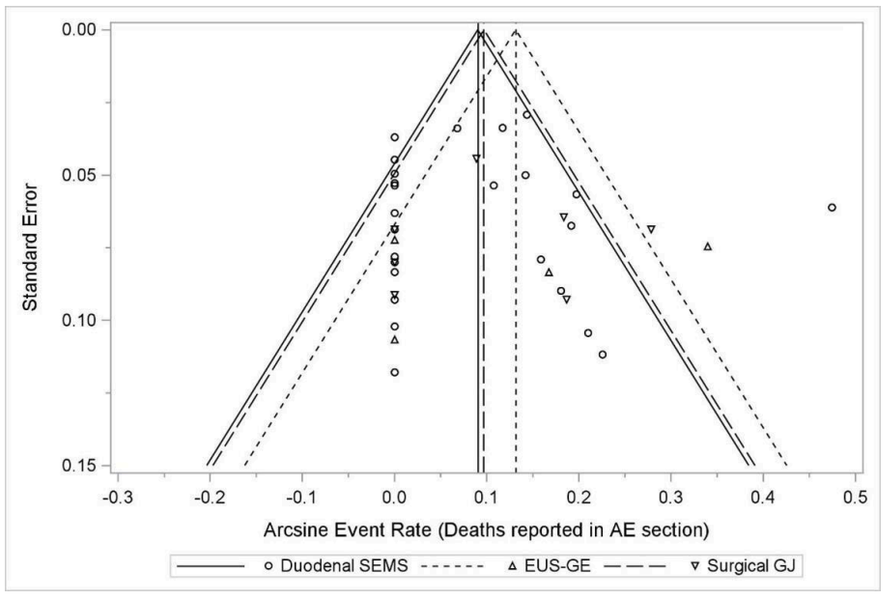
**M.** Deaths reported in AE section



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